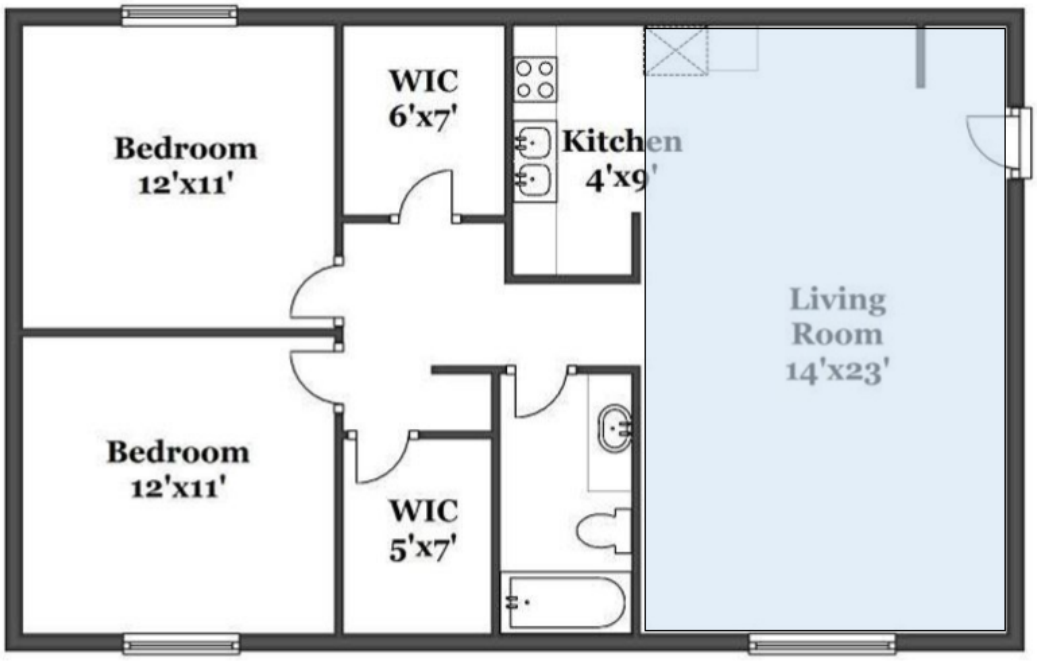


Figure 6.2 A sample floor plan for the toy example of finding the remote control. The living room is uniformly shaded to indicate that the distribution of the location of the remote is uniform over the living room.

Figure 6.2 visualizes this belief about the location of the remote control that you have, indicated by the shaded living room, which is where the remote is (according to your belief). Now, the question you might ask yourself is: where in this house should you look for the remote? It's reasonable to think that you should look in the living room, as opposed to say the bathroom, because that's where the remote is. But how does one quantifiably justify that choice?

Information theory, specifically entropy, offers a way of doing that, by allowing us to reason about how much entropy remains after a search for a remote in the living room vs. one in the bathroom, that is, how much uncertainty about the location of the remote you have left after looking in the living room vs. after looking in the bathroom.

Figure 6.3 shows how entropy of the location of the remote decreases once the upper portion of the living room is searched.

- If the remote is found within the searched region, then you will no longer have any uncertainty about its location. In other words, the entropy is zero.
- If the remote is not found, then your posterior belief about the location of the remote gets updated to the shaded region in the lower right section. This distribution spans a smaller region than the one in Figure 6.2, so there is less uncertainty (entropy).

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